

HAT-P-32b

R-0259

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_230817 · created 2026-07-17T08:46:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460173.8746 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.053
Transit depth (Rp/Rs)^2	3.0 +/- 1.84 [%]
Orbital Inclination (inc)	84.18 +/- 1.74
Airmass coefficient 1 (a1)	3.06 +/- 2.13
Airmass coefficient 2 (a2)	-0.64 +/- 0.4
Scatter in the residuals of the lightcurve fit is	54.31 %
Best Comparison Star	None
Optimal Aperture	6.53
Optimal Annulus	12.46
Transit Duration (day)	0.0678 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.053 vs 0.1489 (deviation 0.34 σ)
Duration fit vs published	0.0678 d vs 0.1304 d (deviation 7.46 σ)
Mid-time O-C	8.3 min
Depth SNR	2.4
Residual scatter	54.31 %

Light curve

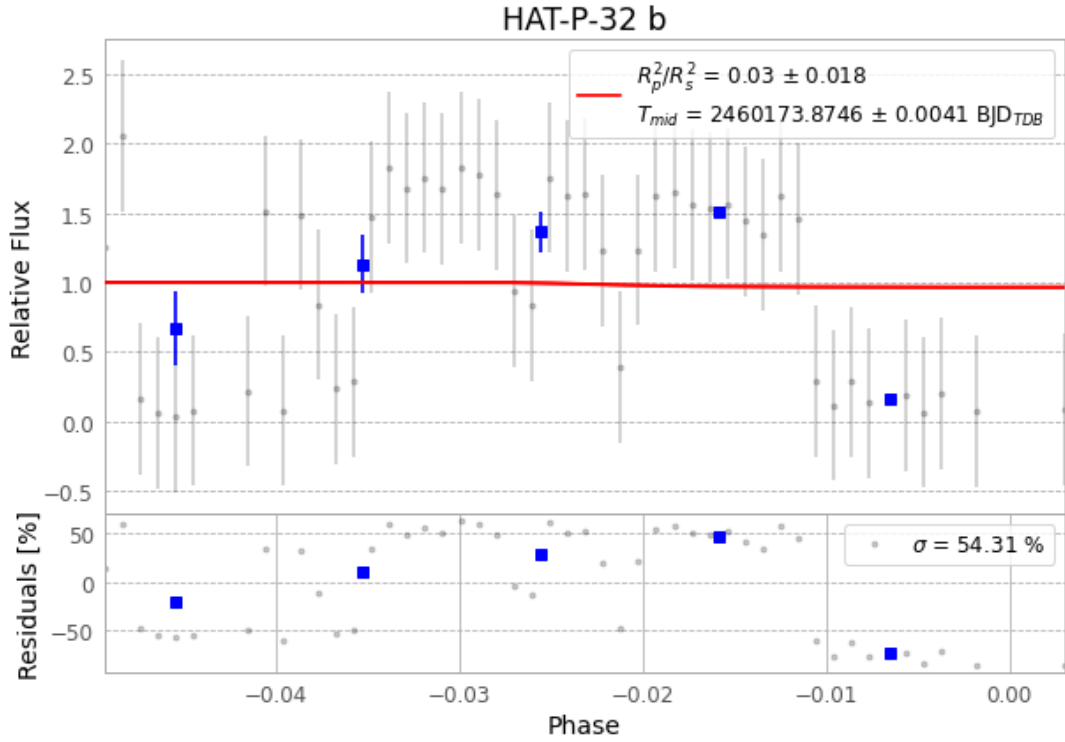


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-08-16.png (SHA-256 3851eecf5018edc9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [14, 41], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa66a86b0087ca0d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 13bb8b5

Data provenance

77 FITS frames (51 MB), HATP-32230817062415.FITS → HATP-32230817101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-230817.log (492b3b0ca182...), FinalLightCurve_HAT-P-32 b_2023-08-16.png (3851eecf5018...), FinalLightCurve_HAT-P-32 b_2023-08-16.csv (8f8deadfb9a5...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 08:46Z.